Human Reproduction

SHORT NOTES

Reproduction

- The male and female reproductive organs work together to produce offspring.
- **Primary sex organs**: Formation of the gametes (sperm and egg) and hormones.
- **Secondary Sex Organs:** Do not produce gametes, but provide passage for the gametes.

Male Reproductive System

- Male reproductive system consists of **primary sex organ** which are testes (one pair) and secondary sex organs which are male accessory ducts which include rete testis, vasa efferentia, epididymis, vas deferens, urethra.
- Male accessory glands include seminal vesicles, prostate gland, Cowper's glands/Bulbourethral gland and male external genitalia is penis.
- Testes are situated outside the abdominal cavity within a pouch called **scrotum,** for the process of spermatogenesis, which provide **2-2.5°C less temperature** than normal body temperature.
- Each testis has **250 compartments** called testicular **lobules**. Each lobule contains one to three seminiferous tubules. Each seminiferous tubules contain 2 types of cells- Sertoli cells (nurse cells) and Leydig cells (interstitial cells) which synthesise and secrete testosterone.
- Sertoli cells function as an endocrine gland, i.e., secrete biochemicals:
  (*i*) Antimullerian hormone
  (*ii*) Inhibin hormone
  (*iii*) Androgen binding protein (ABP)
- The **pathway of sperm** through the male body is:

Seminiferous tubule → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra

Female Reproductive System

- It consists of primary sex organ which are pair of ovaries and secondary sex organs which are oviducts (fallopian tubes), uterus, vagina, female external genitalia and mammary glands.
- **Oviduct:** Infundibulum (funnel-shaped), Ampulla (fertilization takes place) and Isthmus (last part).
- **Uterus:** Corpus/Body, Fundus and Cervix.
- The wall of the uterus has three layers of tissue: perimetrium, myometrium, endometrium.
- Birth canal = Cervical canal + Vagina.
- Female external genitalia consists of mons pubis, labia majora, labia minora, hymen, clitoris.
- Accessory glands in female is of 2 types- greater vestibular or bartholin's gland and lesser vestibular glands or paraurethral or skene's glands.
- Pathway of milk ejection

Glandular tissue → Mammary lobes → Mammary alveoli → Mammary tubules → Mammary duct → Mammary ampulla → Lactiferous duct

Gametogenesis

- The process of formation of gametes. It is divided in to three phases:

1. Multiplication phase
2. Growth phase
3. Maturation phase

Spermatogenesis

- The process of formation of sperms.
- Spermatogonia, Primary spermatocytes, Secondary spermatocytes, Spermatids, Spermatozoa
- **Spermiogenesis:** Transformation of the spermatid into a mature sperm cell, or spermatozoon.
- **Spermiation:** The process of release of sperm from the sperm head into the lumen of seminiferous tubule.
- Sperm is divided in 3 parts- Head, Middle piece (production of energy due to presence of mitochondria) and Tail (help in movement).
- Human male ejaculates about 200–300 million sperms during a coitus of which for normal fertility, at least **60 percent** sperms must have normal **shape and size**, at least **40 percent** of sperm among them must show **vigorous motility**.

Oogenesis

- Process of formation of a mature female gamete. It is initiated during the embryonic development stage with million of gamete mother cells called oogonia. **Tertiary follicle** shows the presence of fluid filled cavity known as **antrum**.
- LH acts on corpus luteum to secrete four hormones- progesterone, oestrogen, relaxin, and inhibin.
- FSH causes the development of ovarian follicles.

Menstrual Cycle (28/29 days)

- The cycle of non-pregnant females in their ovaries and uterus, which involves the periodic shedding of the endometrium. It is divided in 3 phases:

1. Menstrual Phase/Bleeding phase (3-5 days).
2. Preovulatory phase/Proliferative phase/Follicular phase (6 to 13 days).
3. Postovulatory phase/Secretory phase/Luteal phase (**14 days fixed**).

Fertilisation

- The process of fusion of a sperm with an ovum to form a diploid cell is called fertilisation. It involves:

1. Acrosomal reaction
2. Fast block to polyspermy
3. Slow block to polyspermy

Cleavage

- The fertilized egg, undergoes repeated cell divisions which occur rapidly to produce a multicellular structure without changing its size. It involves following stages:

1. Morula 2. Blastula 3. Gastrula

Extra Embryonic Membranes

- There are 4 types of extra embryonic membranes

1. Chorion 2. Yolk sac 3. Allantois 4. Amnion

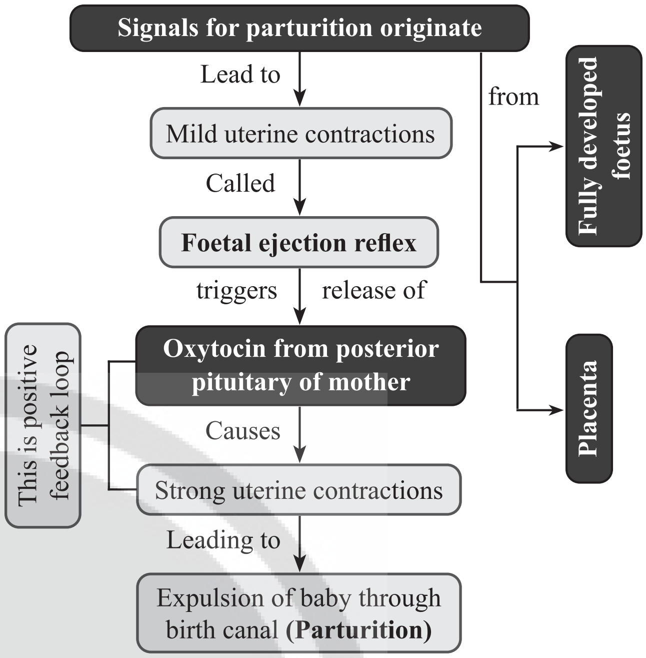
Organogenesis

Major Events During Gestation Period in Humans:

Trimester	Month	Week	Event
1st	I	4	Heart is formed, sign of growing foetus noticed by listening to the heart sounds through stethoscope.
	II (end)	8	Foetus develops limbs and digits.
	III (end)	12	Most of major organ systems are formed including external genital organs, limbs.
2nd	V	20	First movement of foetus, appearance of hair on head.
	VI (end)	24	Body is covered with fine hair, eyelids separate, eyelashes are formed.
3rd	IX (end)	36	Foetus is fully developed and is ready for delivery.

Parturition

- The process of childbirth which is induced by a complex neuroendocrine mechanism.



Lactation

- The process of formation of milk at the end of pregnancy for bringing up a healthy baby.
- **Prolactin** helps in production of milk.
- **Oxytocin** causes ejection of milk.
- In initial few days of lactation **colostrum** (contains IgA) is released by lactating mother.